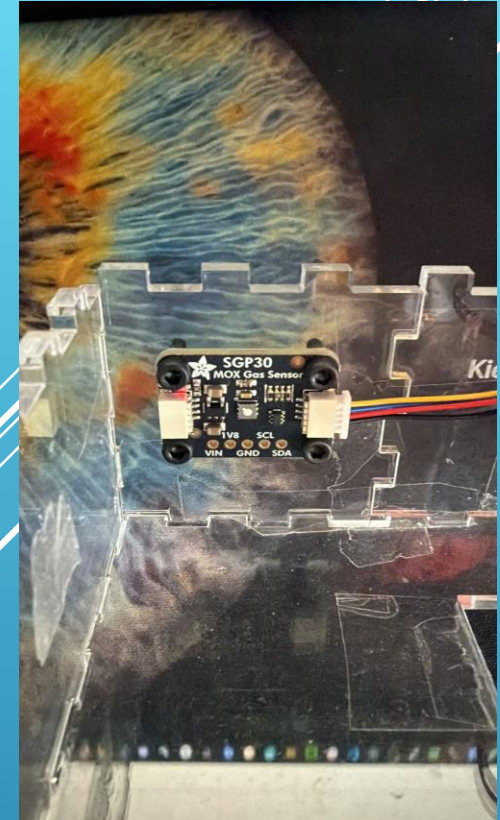


The SGP30 communicates over I2C at address 0x58.

Sensor: SGP30

KIERAN SHARMA

The SGP30 is an air quality sensor that checks how fresh or polluted the air is. It estimates carbon dioxide levels and detects chemical vapors or odors, helping you know if the air needs ventilation.



```
pi@raspberrypi:~$ sudo i2cdetect -y 1
   0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00: -- -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- 1c -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- 39 -- -- -- -- --
40: -- -- -- -- -- UU -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- 58 -- -- -- 5c -- -- 5f
60: -- -- -- -- -- -- -- 6a -- -- -- -- -- --
70: -- -- -- -- -- -- -- -- -- -- -- -- -- --
pi@raspberrypi:~$
```

```
pi@raspberrypi:~/CENG317/Test$ python3 sgp30_readings.py
SGP30 Air Quality Sensor Example
-----
Waiting 15 seconds for sensor to stabilize...
eCO2: 400 ppm (Excellent - fresh air) | TVOC: 0 ppb (Excellent - very clean air)
eCO2: 400 ppm (Excellent - fresh air) | TVOC: 0 ppb (Excellent - very clean air)
eCO2: 400 ppm (Excellent - fresh air) | TVOC: 0 ppb (Excellent - very clean air)
eCO2: 3834 ppm (Very Poor - ventilation needed!) | TVOC: 192 ppb (Good - typical indoor air)
eCO2: 2163 ppm (Very Poor - ventilation needed!) | TVOC: 357 ppb (Moderate - some pollutants detected)
eCO2: 695 ppm (Good - normal indoor air) | TVOC: 234 ppb (Moderate - some pollutants detected)
eCO2: 514 ppm (Excellent - fresh air) | TVOC: 159 ppb (Good - typical indoor air)
eCO2: 459 ppm (Excellent - fresh air) | TVOC: 112 ppb (Good - typical indoor air)
eCO2: 645 ppm (Good - normal indoor air) | TVOC: 95 ppb (Good - typical indoor air)
eCO2: 584 ppm (Excellent - fresh air) | TVOC: 80 ppb (Good - typical indoor air)
eCO2: 505 ppm (Excellent - fresh air) | TVOC: 57 ppb (Excellent - very clean air)
eCO2: 479 ppm (Excellent - fresh air) | TVOC: 46 ppb (Excellent - very clean air)
eCO2: 435 ppm (Excellent - fresh air) | TVOC: 31 ppb (Excellent - very clean air)
```